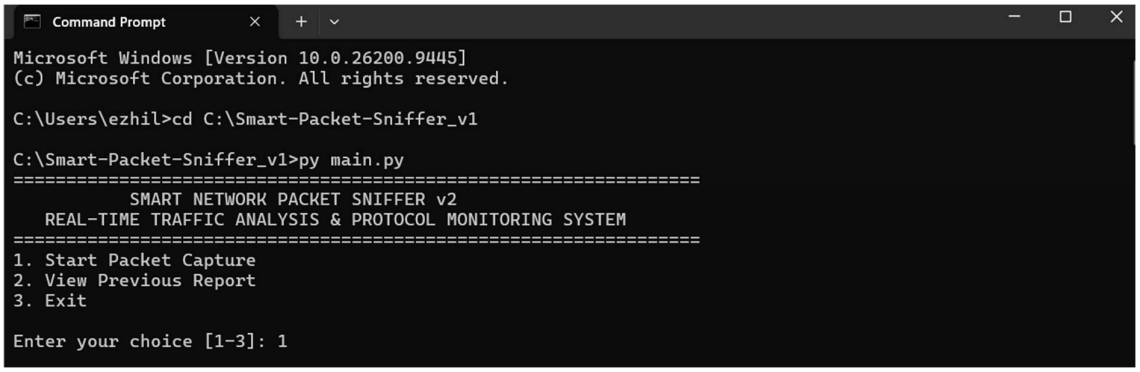
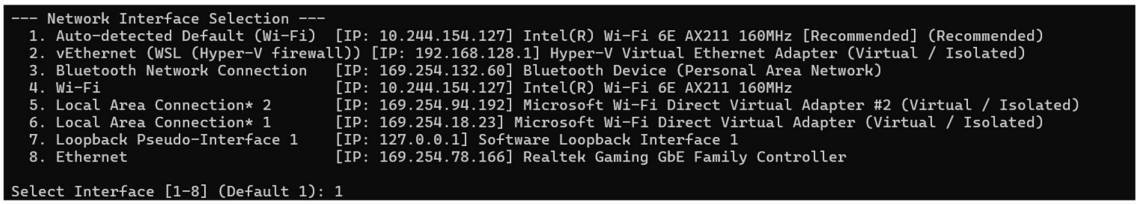

Command Prompt
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.
C:\Users\ezhil>cd C:\Smart-Packet-Sniffer_v1
C:\Smart-Packet-Sniffer_v1>py main.py
SMART NETWORK PACKET SNIFFER v2
REAL-TIME TRAFFIC ANALYSIS & PROTOCOL MONITORING SYSTEM
1. Start Packet Capture
2. View Previous Report
3. Exit
Enter your choice [1-3]: 1


--- Network Interface Selection ---
1. Auto-detected Default (Wi-Fi) [IP: 10.244.154.127] Intel(R) Wi-Fi 6E AX211 160MHz [Recommended] (Recommended)
2. vEthernet (WSL (Hyper-V firewall)) [IP: 192.168.128.1] Hyper-V Virtual Ethernet Adapter (Virtual / Isolated)
3. Bluetooth Network Connection [IP: 169.254.132.60] Bluetooth Device (Personal Area Network)
4. Wi-Fi [IP: 10.244.154.127] Intel(R) Wi-Fi 6E AX211 160MHz
5. Local Area Connection* 2 [IP: 169.254.94.192] Microsoft Wi-Fi Direct Virtual Adapter #2 (Virtual / Isolated)
6. Local Area Connection* 1 [IP: 169.254.18.23] Microsoft Wi-Fi Direct Virtual Adapter (Virtual / Isolated)
7. Loopback Pseudo-Interface 1 [IP: 127.0.0.1] Software Loopback Interface 1
8. Ethernet [IP: 169.254.78.166] Realtek Gaming GbE Family Controller
Select Interface [1-8] (Default 1): 1


[+] Selected Interface: Auto-detected Default (Wi-Fi) (10.244.154.127)
--- Choose Capture Mode ---
1. Fixed Packet Count
2. Continuous Capture (Manual Stop with Ctrl+C)
Enter choice [1-2] (Default 1): 1
Enter number of packets to capture [1-10000] (Default 20): 25


--- Choose Protocol Filter ---
1. All Packets (No Filter)
2. TCP Packets Only
3. UDP Packets Only
4. ICMP Packets Only
5. ARP Packets Only
6. Custom BPF Filter Expression
Enter choice [1-6] (Default 1): 1


STARTING CAPTURE SESSION : 20261012_224201
Interface : Auto-detected Default (Wi-Fi)
Mode : Fixed (25 packets)
Filter : All Traffic
Press Ctrl+C at any time to stop capture gracefully.

```
----------------------------------------------------------------------
PACKET #1     | Time: 22:42:01 | Type: IPv4
----------------------------------------------------------------------
Source      : 57.144.55.201        (Port: 40007)
Destination : 10.244.154.127       (Port: 49666)
Protocol    : UDP
Service     : Other / Dynamic (Common Service Mapping)
Length      : 1183 Bytes
----------------------------------------------------------------------

----------------------------------------------------------------------
PACKET #2     | Time: 22:42:01 | Type: IPv4
----------------------------------------------------------------------
Source      : 10.244.154.127       (Port: 49666)
Destination : 57.144.55.201        (Port: 40007)
Protocol    : UDP
Service     : Other / Dynamic (Common Service Mapping)
Length      : 84 Bytes
----------------------------------------------------------------------

----------------------------------------------------------------------
PACKET #3     | Time: 22:42:01 | Type: IPv4
----------------------------------------------------------------------
Source      : 57.144.55.201        (Port: 40007)
Destination : 10.244.154.127       (Port: 49666)
Protocol    : UDP
Service     : Other / Dynamic (Common Service Mapping)
Length      : 1183 Bytes
----------------------------------------------------------------------

----------------------------------------------------------------------
PACKET #4     | Time: 22:42:01 | Type: IPv4
----------------------------------------------------------------------
Source      : 10.244.154.127       (Port: 49666)
Destination : 57.144.55.201        (Port: 40007)
Protocol    : UDP
Service     : Other / Dynamic (Common Service Mapping)
Length      : 1127 Bytes
----------------------------------------------------------------------
```

```
================================================================
                  TRAFFIC SUMMARY & ANALYTICS
================================================================
Session ID           : 20261012_224201
Capture Duration     : 0.10 seconds
Total Packets        : 25
Total Data           : 23,209 Bytes (22.67 KB)
Packet Throughput    : 247.81 pkts/sec
Data Throughput      : 230058.57 B/s (1840.47 Kbps)
Average Packet Size  : 928.36 Bytes
Min / Max Size       : 84 / 1184 Bytes
IP / Non-IP Frames   : 25 IP / 0 Non-IP
----------------------------------------------------------------
PROTOCOL DISTRIBUTION:
  TCP   : 0      (  0.0 %)
  UDP   : 25     (100.0 %)
  ICMP  : 0      (  0.0 %)
  ARP   : 0      (  0.0 %)
  Other : 0      (  0.0 %)
----------------------------------------------------------------
TOP SOURCE IP ADDRESSES:
  - 10.244.154.127            :    17 pkts (  15,047 B)
  - 57.144.55.201             :     8 pkts (   8,162 B)
----------------------------------------------------------------
BASIC RULE-BASED TRAFFIC ANOMALY DETECTION:
Anomaly Status : NORMAL
  [+] No rule-based anomalies detected.
================================================================

[+] Packet log saved to : logs\session_20261012_224201.csv
[+] Traffic report saved to: reports\report_20261012_224201.txt
================================================================
```

```
================================================================
          SMART NETWORK PACKET SNIFFER v2
   REAL-TIME TRAFFIC ANALYSIS & PROTOCOL MONITORING SYSTEM
================================================================
1. Start Packet Capture
2. View Previous Report
3. Exit

Enter your choice [1-3]: 3

Exiting Smart Network Packet Sniffer. Goodbye!
```